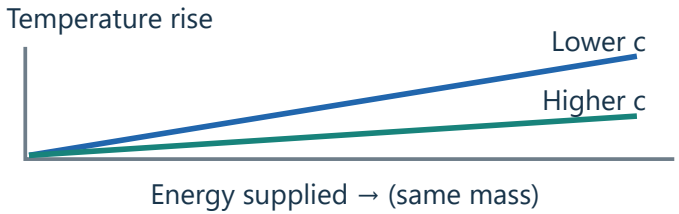
Heating & practicals

Link energy & temperature

1 Specific heat capacity



Energy to warm **1 kg by 1°C** without a state change. Units: J/kg°C.

ΔE = mcΔθ. Higher c → less warming for the same mass and input.

2 Worked example

0.50 kg water, c = 4,200 J/kg°C, warmed 20 → 40°C.

Δθ = 20°C; ΔE = 0.50 × 4,200 × 20 = **42,000 J.**

Convert grams to kg first.

Measure fairly

3 Measure c

Balance → mass; thermometer → rise; joulemeter or Pt → energy.

Insulate a block; improve heater/probe contact. Record initial temperature, heat, record rise.

c = E/(mΔθ). Repeat; keep electrical parts dry; avoid hot surfaces.

4 Explain error

Energy also heats equipment and surroundings. Using all input for the sample makes c too high.

Insulation reduces losses. Good thermal contact reduces lag. Energy vs temperature-rise gradient = mc ideally.

Limit cooling

5 Insulators

Thicker material or lower conductivity → slower conduction under matching conditions.

Trapped air reduces transfer; draught-proofing limits escaping warm air.

Payback time = cost / yearly saving, assuming constant savings.

6 Cooling investigation

Compare equal thicknesses; same containers, lids, water mass, starting temperature and room.

Measure temperature at equal intervals; repeat and average. Smaller drop in same time → better insulation.

For thickness: keep material fixed. Handle hot water safely.